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The influence of supply network structure on firm innovation

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ABSTRACT

In this study, we examine the structural characteristics of supply networks and investigate the relationship between a firm's supply network accessibility and interconnectedness and its innovation output. We also examine potential moderating effects of absorptive capacity and supply network partner innovativeness on innovation output. We hypothesize that firms will experience greater innovation output from (1) higher levels of supply network accessibility and supply network interconnectedness, (2) the interaction between the levels of these two structural characteristics, (3) the moderating role of absorptive capacity on supply network accessibility and the moderating role of supply network partner innovativeness on supply network interconnectedness. Supply network partner relationships are drawn in the context of the electronics industry using data from multiple sources. We use social network analysis to create measures for each supply network structural characteristic. Using regression techniques to test the relationship between these structural characteristics and firm innovation for a sample of 390 firms, our findings suggest that supply network accessibility has a significant association with a firm's innovation output. The results also indicate that interconnected supply networks strengthen the association between supply network accessibility and innovation output. Moreover, the influence of the two structural characteristics on innovation output can be enhanced by a firm's absorptive capacity and level of supply network partner innovativeness. By addressing the need for deeper structural analysis, this study contributes to supply chain research by accounting for the embedded nature of ties in supply networks, and showing how these structural characteristics influence the knowledge and information flows residing within a firm's supply network.

1. Introduction

Supply chains manifest networks that are comprised of not only a focal firm's direct ties to each of its supply network partners (e.g., suppliers and customers), but also its indirect ties to partners of the firm's direct partners (Choi et al., 2001). Both anecdotal evidence and research on supply networks highlight the operational benefits of effectively managing a supply network. Firms such as Toyota and Schneider Electric have been creating and reevaluating their supply networks to maintain efficiencies in inventory, improve product quality, enhance delivery performance, mitigate supply chain disruptions and enhance profitability (Dyer and Hatch, 2004; Voxant FD Wire, 2009).

Besides these operational benefits, the supply network of a firm has also been viewed as a source of innovation. For example, Procter and Gamble (P&G) has set an imperative of sourcing innovation from outside the firm. P&G's CEO, A.G. Lafley, confirmed this imperative in 2002 by stating "we will acquire 50% of our technologies and products from outside P&G" (Huston and Sakkab, 2006). These sources included amongst others, consumers, universities and suppliers. Examples are also aplenty in knowledge-intensive industries such as electronics. For instance, the CEO of Direct Methanol Fuel Cell Corporation (DMFCC) announced that the firm "has been establishing a global network of suppliers to manufacture fuel cartridges and other fuel cell products" (PR Newswire, 2007). In particular, the announcement cites the role of its supplier, Tyco Electronics Corporation, in developing and commercializing the innovation on fuel cell technology. Similarly, in 2012, Dell, with the help of its reverse logistics provider GENCO, has initiated innovation in products and processes with several of its suppliers (Gilmore, 2012). In

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line with these examples, research has also conceptualized a firm's partners as sources of innovation and empirically examined the role of partner integration into innovation and new product development activities (Azadegan et al., 2008; Choi and Krause, 2006; Dyer and Nobeoka, 2000; Von Hippel, 1988).

However, little is known about the underlying structural characteristics of a firm's supply network and whether these characteristics have any influence on firm innovation output. Several supply chain researchers have emphasized the value in incorporating network structure when considering firm innovation and performance implications (Autry and Griffis, 2008; Bernardes, 2010; Choi and Kim, 2008). A recent research note by Narasimhan and Narayanan (2013) discusses the role of structural characteristics of a firm's supply network on innovation. The note also emphasizes the role of absorptive capacity, in integrating information flows from supply network partners, to facilitate innovation output. Specifically, absorptive capacity reflects a firm's ability to recognize, assimilate, leverage, and deploy the available external knowledge (Cohen and Levinthal, 1990). Moreover, the emerging paradigm of open innovation suggests that innovations are also derived outside a firm's internal endeavors, and that greater, more novel learning is often gained from external sources (Chesbrough, 2003). Accordingly, firms are recognizing the advantages of leveraging the innovativeness of their supply network partners to influence their innovation output. Specifically, *supply network partner innovativeness* reflects the magnitude of available knowledge residing in a firm's supply network partners (Azadegan et al., 2008).

Our study builds on the conceptualization presented in extant research and empirically addresses two interrelated research questions: *First*, w*hat is the association between the structure of a firm's supply network and its innovation output?* Specifically, we examine two important structural characteristics of supply networks: *supply network accessibility* – the speed and effectiveness of information and knowledge access opportunities between a firm and its supply network – and *supply network interconnectedness* – the extent to which a firm's supply network partners are inter-linked. *Second, what moderating role does a firm's absorptive capacity and its supply network partner innovativeness play in the association between the structural characteristics of a firm's supply network and its innovation output?* To empirically test the hypothesized relationships, we collected firm-level data from several archival sources that included data on buyer–supplier relationships, alliances, and patenting activity and used social network analysis to develop the structural characteristics.

We contribute to the literature on examining a firm's supply network partners as a source of innovation in the following ways. First, a firm's level of innovation output is a by-product of its knowledge creation activities and often results in inventions and commercialization that reflect advancements over existing technology or practices. Previous research recognizes a firm's partners as sources of innovation, with firms tapping into the knowledge of their partners (Dyer and Nobeoka, 2000; Von Hippel, 1988). Further, other scholars have argued the need for future research using a more comprehensive structural analysis that accounts for the embedded nature of knowledge among members in the supply network (Kim et al., 2011; Villena et al., 2011). Moreover, prior research has conjectured that the way a firm's supply network is structured, formally termed as structural characteristics, will bear influence on its innovation performance, and have therefore called for future research to empirically examine and test such conjectures (e.g., Autry and Griffis, 2008). We extend these stances by specifically examining the structural characteristics of a firm's supply network and its association with innovation performance by taking a more holistic view of firms that are embedded in a supply network, rather than the traditional dyadic view. In this regard, we consider the influence of two key structural characteristics in a supply network on innovation: supply network accessibility and supply network interconnectedness.

Second, in addition to addressing the influence of structural characteristics of a supply network on innovation, we also examine the moderating role of two critical knowledge variables, the presence of which may strengthen the relationship between the two structural characteristics and innovation output. We examine the moderating role of *absorptive capacity*, represented by the firm's research and development (R&D) intensity. We also consider the role of *supply network partner innovativeness*, that is, the magnitude of available knowledge or more formally, the average level of patent stock that exists among a focal firm's supply network partners. While the structural characteristics of a supply network facilitate knowledge and information flows between partners, both knowledge availability in the network and the capability to combine the knowledge may enable firms to translate the benefits of high levels of supply network accessibility and supply network interconnectedness into higher innovation output. Our findings confirm this conjecture, suggesting that opportunity exists for firms in knowledge-intensive industries such as electronics – which outsource parts of their product development to supply network partners – to supplement growth in their innovation output through leveraging its absorptive capacity and the innovativeness of its supply network partners (Dedrick et al., 2010). Overall, taking the main effects of structural characteristics together with the moderating effects of knowledge variables provides a coherent theoretical framework to potentially extend the literature on the influence of the structural characteristics of supply networks on innovation.

The remainder of this study is organized as follows. In Section 2, we discuss the literature on innovation, supply chain management, and networks and develop hypotheses relating supply network structural characteristics to firm innovation output. We describe our research methodology in Section 3 and our empirical analysis and results in Section 4. In Section 5 we discuss our research findings, implications for theory and practitioners, and future research opportunities. Lastly, we conclude our study in Section 6.

2. Theoretical background and hypotheses development

Buyer–supplier relationships have been conventionally viewed as linear or dyadic structures, rather than as a network (Kim et al., 2011). However, given a supply chain's complex and increasingly interdependent nature, a network approach provides a richer view by considering the various interactions taking place among firms in the supply network (Borgatti and Li, 2009; Buhman et al., 2005; Choi et al., 2001). In contrast to the conventional approach, where firms are viewed as autonomous and self-reliant entities striving to use their resources to compete with other such entities, the network approach focuses on the structural elements of the firm and its inter-organizational network partners (Granovetter, 1985). We define a supply network as an inter-linked network of firms consisting of manufacturers, suppliers, customers, third party service providers, and alliance partners that interact to execute the supply chain activities of the firm. The various firms in the supply network are generally referred to as supply network partners of a given focal firm in the network.

The theory on social networks helps explain the benefits derived by a firm viewed as embedded within a larger network of structurally interdependent partners. This lens emphasizes that the benefits accrued from access to knowledge, resources, and information available within a network of relationships can lead to an organizational advantage (Granovetter, 1973). Nahapiet and Ghoshal (1998) elaborate on this notion across three dimensions: cognitive, relational, and structural. The cognitive dimension refers to those resources that help generate shared language and vocabulary and the sharing of collective narratives (Nahapiet and Ghoshal, 1998). Collective goals and aspirations between partners can thus enhance the cognitive dimension and help stifle opportunistic behavior and improve joint returns for both parties (Villena et al., 2011). The relational dimension refers to the degree of mutual respect, trust, and close interaction that exists between a firm and its partners (Granovetter, 1992; Kale et al., 2000). From a relational view, firms can profit from collaborative efforts with supply network partners by creating joint benefits that may have not been possible to create by either firm in isolation (Dyer and Singh, 1998). Lastly, the structural dimension refers to the overall pattern of connections between partnering firms, mapping who a particular firm reaches and how they reach them. From this dimension, the network structure derived from a firm's compendium of ties determine, in part, opportunities and constraints to access valuable resources and information that would help them sustain a competitive advantage (Burt, 1992). As mentioned earlier, in this study we focus on the structural dimension, and investigate the influence of two key *structural* characteristics of a firm's supply network on its innovation output – supply network accessibility and supply network interconnectedness – to examine the role of supply network structure as a source of innovation.

2.1. Supply network as a source of innovation

Innovation acts as an enabler to develop unique products and services that help a firm gain competitive advantage. It is well established in the literature that the accrual of knowledge assets that drive innovation in firms come from two primary sources: internal knowledge generation and knowledge derived from external sources. Generating internal knowledge assets can come from, for example, the progression of technical systems for effective experimentation, prototyping, simulation, and testing during product or service development (Gaimon and Bailey, 2012; Thomke, 1998). Deriving knowledge from external sources is part of vicarious learning, whereby organizations acquire knowledge and experience from other external organizations (Dutton and Freedman, 1985; Hora and Klassen, 2013). Past literature in organizational learning has also emphasized the potential knowledge gain from external sources, such as suppliers, customers, service providers, and alliance partners in a firm's supply network (Yli-Renko et al., 2001). Thus, supply networks serve as important conduits and sources of information and knowledge access, and act as catalysts for the development and dissemination of new ideas, applications, and supply chain practices. This organizational phenomenon is espoused by social network theorists asserting that innovation performance advantage can be accrued by the knowledge and information assets derived from the structural linkages among firms in a network.

A core principle of the creation of the structural dimension in the social network is that firms are embedded in a larger network of supply partners, comprising of other firms that are able to provide access to unique resources, information, and influence (Granovetter, 1973). In that light, supply networks not only contain dyadic relationships between partners but also act as critical conduits of knowledge and information flow. Firms can leverage this dimension to facilitate the knowledge flows by influencing the conditions necessary for exchange and combination amongst its partners to occur (Nahapiet and Ghoshal, 1998). Several studies have underlined the benefits of structural characteristics such as a firm's structural position in its supply network (e.g., Burt, 2001; Kim et al., 2011; Yli-Renko et al., 2001; Zaheer and Bell, 2005).

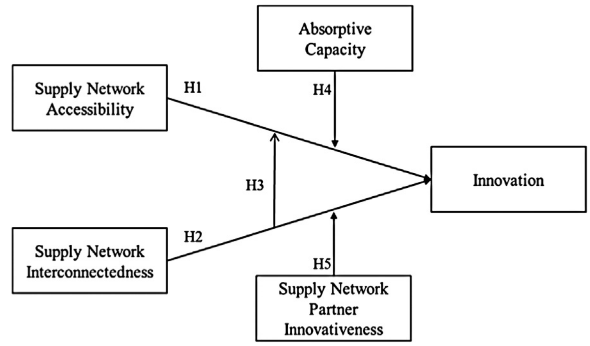

Fig. 1. Conceptual model.

2.2. Structural characteristics of supply networks

From the social network perspective, a network consists of entities (e.g., individuals, groups, firms) represented as nodes and ties between them represented as links (Wasserman and Faust, 1994). In the supply network context, the entities reflect suppliers, manufacturers, service providers, alliance partners, and customers that are linked together through activities related to the procurement and transformation of raw materials in order to produce and deliver goods and services (Kouvelis et al., 2006; Lamming et al., 2000). Two important structural characteristics that may act as important enablers of the flow of information and knowledge in the network are (i) how effectively a firm is able to access the different sources of information and knowledge assets in the network, formally termed *supply network accessibility*, and (ii) how these sources of information and knowledge are structurally inter-linked together in the network, formally termed *supply network interconnectedness*. We posit that these two structural characteristics of a supply network have significant influence on the innovation output of firms. These relationships are depicted in Fig. 1, and discussed in more detail in the ensuing sections.

2.2.1. Supply network accessibility

Supply network accessibility refers to the effectiveness with which a firm can access information and knowledge from other members in its supply network, including indirect access to members with whom they do not share a direct relationship with. In addition, it also reflects the speed of information access. While the formal measure of this structural characteristic is given in Section 3.3.1, the ease and effectiveness of information and knowledge access by a focal firm from members in the supply network is influenced by the distance between each member and the focal firm in terms of the mean length of the path between them. In other words, high (low) levels of supply network accessibility allow a firm to traverse fewer (more) steps or connecting points to reach supply network members, for example, lower tier suppliers. Fig. 2 illustrates the different levels of accessibility that a firm can have in its supply network.

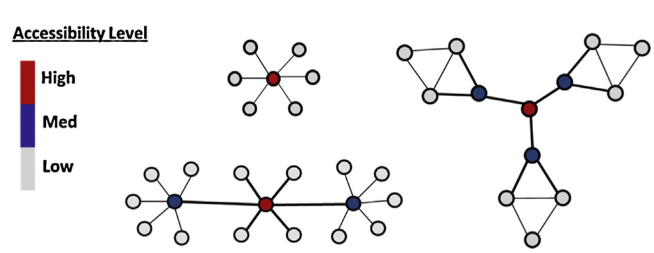

Fig. 2. Illustration of firms with different levels of supply network accessibility.

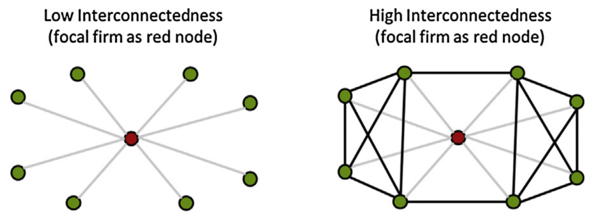

Fig. 3. Illustration of a lowly versus highly interconnected supply network.

Certain relationships in a firm's supply network – both direct and indirect – may be more suited to foster innovation. Because of their unique structural position within the network, some firms may be able to avail high levels of supply network accessibility and can thus access and transmit knowledge and information across the supply network faster. Such firms are able to reach a large number of members through a fewer number of intermediate connecting points (i.e., partners, other members) and are better positioned to obtain information quickly and with reduced risk of information distortion (Schilling and Phelps, 2007). Firms that have this favorable structural position in the supply network are referred to as central firms (Kim et al., 2011), and are reflected in Fig. 2 as the darker nodes.

While prior studies have shown several operational benefits that derive from information access and transmission among supply chain members – such as reduced supply chain costs, shorter lead times and smaller batch sizes (Cachon and Fisher, 2000), lower inventory holding and shortage costs (Lee et al., 2000), and reduced stockouts (Kulp et al., 2004) – there are also other benefits which lead to an increase in a firm's innovation output. Thus, firms with high levels of supply network accessibility may have access to more opportunities to obtain novel information or, in the case of R&D technology-sharing, to develop products sooner than their competitors (Borgatti and Molina, 2005). This information-based advantage becomes more critical in the context of fast-moving industries characterized by uncertain demand, low product life cycles, and highly innovative products (Kanda and Deshmukh, 2008).

Therefore, we argue that possessing higher levels of supply network accessibility allows a firm wider reach and access to knowledge and information in the network, which will enhance their potential to receive knowledge spillovers faster than others, and thus increases the likelihood of higher innovation output.

Hypothesis 1. The level of accessibility in a firm's supply network is positively associated with its innovation output.

2.2.2. Supply network interconnectedness

While supply network accessibility reflects the effectiveness with which a firm can access the knowledge and information sources in the supply network at large, supply network interconnectedness reflects the potential sources or ports of knowledge that reside because of the shared linkages among a firm's direct partners. More formally, supply network interconnectedness refers to the degree to which supply network partners of a focal firm are connected to each other, and thus share direct links amongst themselves. Supply networks are considered to be densely interconnected when there are a large number of shared linkages that exist between the supply network partners of a focal firm. The notion of supply network interconnectedness can be seen through the simplified illustration shown in Fig. 3. These illustrations depict two companies with similar supply network size, but with different levels of supply network interconnectedness. The depiction on the left of Fig. 3 shows a focal firm with all of its direct partners only connected to it and no other source, leading to a network with low interconnectedness. Conversely, the depiction on the right shows a focal firm with the same number of direct partners, but this time where each partner shares at least three direct links with the other remaining partners, thus increasing the level of interconnectedness of the firm's supply network.

We posit that increased interconnectedness in a supply network is expected to positively influence innovation output for the following reasons. Inkpen and Tsang (2005, p. 152) suggest that interconnectedness through multiple knowledge connections enables "ease of knowledge exchange" for a focal firm and can thus enhance the flow of information and knowledge among its members. These benefits from increased interconnectedness in a supply network are derived due to the potential for richer collaboration, resource pooling, and problem solving within structurally embedded, dense cliques (Ahuja, 2000; Schilling and Phelps, 2007). Lavie (2006) conceptualized such benefits as inbound spillovers for the focal firm which may provide a competitive advantage. In addition, a highly interconnected network allows a focal firm to alleviate appropriation concerns and consider its partners more trustworthy (Echols and Tsai, 2005). Accordingly, collaborative environments within the supply network can facilitate the sharing of knowledge, enhance knowledge creation, and increase innovation activities in general (Inkpen, 1996; Simatupang and Sridharan, 2002). Moreover, the series of redundant ties – where a focal firm has multiple indirect ties to the same partner through more than one direct relationship – also enables the focal firm to validate the reliability of information that is exchanged in its supply network. Taken together "interconnected ties provide a firm with the kind of highly reliable and trustworthy partnerships needed to protect and advance its special knowledge" (Echols and Tsai, 2005, p. 223). Thus, in the context of supply networks, we propose that high levels of interconnectedness provide the focal firm, through the fostering of a more collaborative environment, a greater opportunity for accruing the benefits of knowledge spillovers in its innovation output.

Hypothesis 2. The level of interconnectedness in a firm's supply network is positively associated with its innovation output.

2.2.3. Interaction between supply network accessibility and interconnectedness

In our first two hypotheses, we argued that knowledge and information flow benefits are expected from both supply network interconnectedness and supply network accessibility. Building on the basis that both supply network structural characteristics can lead to knowledge and information flow improvements in their own respect, we also posit a positive effect of the interaction of supply network interconnectedness and supply network accessibility on firm innovation output. In particular, firms operating in supply networks that possess both high both potential for collaboration enabled by many direct partners being inter-linked, and quicker ability to reach and access to knowledge and information in the network by accessibility, should experience greater innovation output.

Lower levels of interconnectedness between a focal firm's supply network partners may lead to lower levels of trust and a higher threat of opportunistic behavior, and hence lower resource-sharing benefits, potentially hindering innovation gains from relationships established. While firms with high supply network interconnectedness have a higher concentration of shared ties among its direct partners, a lack of connectivity in the wider network of a given industry can be exploited by boundary-spanning firms that cultivate more opportunities to increase their level of access to diverse information across the wider supply network (i.e., increase their supply network accessibility). Firms that span these clusters often occupy positions of considerable influence (Provan et al., 2007), suggesting that firm efforts to increase their level of supply network interconnectedness and supply network accessibility in tandem should further enhance their ability to positively influence innovation output. In sum, we predict an interaction effect between supply network interconnectedness and supply network accessibility in their effect on knowledge and information flow benefits to the focal firm.

Thus, we posit that firms that maintain highly interconnected supply networks while having higher levels of supply network accessibility will experience greater knowledge and information access and sharing, which is expected to have a positive effect on innovation output.

Hypothesis 3. The interconnectedness of a firm's supply network positively moderates the association between its supply network accessibility and innovation output.

2.3. Moderating roles of knowledge variables on innovation output

In addition to the structural characteristics of a supply network developed above, we also examine the moderating influence of two important knowledge variables on the innovation output of firms. Specifically, we examine the moderating role of a firm's *absorptive capacity*, represented by the firm's ability to recognize, assimilate, and leverage knowledge, as well as the role of *supply network partner innovativeness*, reflecting the magnitude of knowledge available in the firms constituting the supply network. While the structural characteristics of a supply network reflects the effectiveness of how knowledge and information can be accessed as well as shared among the network partners, the knowledge variables – absorptive capacity and level of supply network partner innovativeness – are expected to beneficially influence innovation output. This provides the opportunity to investigate a more coherent theoretical framework, whereby the moderating effect of the knowledge variables can be ascertained in addition to the main and interaction effects of the structural variables on the innovation output of firms in a supply network.

2.3.1. Moderating role of absorptive capacity

While supply network accessibility can lead to greater access to information from supply network members, we further posit that focal firms that can absorb and internalize this available information and knowledge will gain additional benefits related to its innovation performance. The extent of absorbing this knowledge is termed as absorptive capacity, and is defined as a firm's ability to understand, assimilate, and deploy knowledge obtained from other firms, and leverage this knowledge to their benefit (Cohen and Levinthal, 1990; Yli-Renko et al., 2001). A focal firm's access to knowledge and information flow from its supply network members may provide a diverse set of ideas, expertise, and capabilities (Deeds and Decarolis, 1999). This heterogenous knowledge base in the supply network has the potential of resulting in greater innovation for the focal firm (Tsai, 2001). However, the absorptive capacity of a firm influences its ability to adapt the available and accessible external information for its own needs for knowledge creation (Weigelt and Sarkar, 2009).

Ernst and Kim (2002) argue that the combination of both accessibility of information in the network and absorptive capacity are important for a focal firm to develop its innovation capabilities from external knowledge. Furthermore, Easterby-Smith et al. (2008, p. 679) suggest that "absorptive capacity and intra-organizational transfer capability are interrelated in the sense that an organization which is good at absorbing external knowledge should also be well equipped for diffusing the knowledge within its own boundary". In other words, absorptive capacity enables a focal firm to absorb available and accessible external information and knowledge from its partners in the supply network, and ensure that it can be leveraged for its own knowledge creation. In the absence of high absorptive capacity, a firm can still benefit from having access to knowledge in its supply network, but its ability to leverage this information to improve its innovation performance will be very limited. Therefore, we posit a moderating effect of a focal firm's absorptive capacity on the relationship between its supply network accessibility and its innovation output.

Hypothesis 4. A firm's absorptive capacity positively moderates the association between its supply network accessibility and innovation output.

2.3.2. Moderating role of supply network partner innovativeness

Supply network partner innovativeness refers to the level of technological know-how, unique knowledge, and other innovation related capabilities a network partner has accumulated over time. Insights into each potential partner's level of innovativeness can offer an indication of the magnitude of available knowledge that can spill over to other firms in the supply network, and consequently benefit a firm's future innovation-based activities. Azadegan et al. (2008) suggest that supplier innovativeness not only provides the buying firm with improved manufacturing capabilities due to the embedded nature of the supplied component, but also key learning from its suppliers in the process.

We argue that higher levels of innovativeness among the members in a supply network also enhance a firm's potential to benefit from the higher quality knowledge spillovers, and thus provide novel ways to recombine and leverage knowledge, problems, and solutions. Thus, while operating in a highly interconnected supply network can help facilitate knowledge flow and sharing opportunities among partners, the magnitude of knowledge available from a firm's partners can significantly benefit its innovation output. Consequently, this effect should trickle down to help enable the focal firm to develop new technology in its future endeavors (Stuart, 2000). Thus, we conjecture that the existence and level of external knowledge available to a focal firm via the innovativeness of its supply network partners, beneficially moderates the influence of the firm's supply network interconnectedness on its innovation output.

Hypothesis 5. The innovativeness of a firm's supply network partners positively moderates the association between its supply network interconnectedness and innovation output.

3. Methods

In this section, we describe our data sources, steps in constructing the dataset, operationalization of variables and model specification.

3.1. Research setting and data collection

We constructed a database from the following data sources: the Electronics Business 300 (EB 300) listings, the Connexiti database, and the Thomson Reuters SDC Platinum Joint Ventures/Alliances (SDC) database. Our data consisted of active supplier, customer, and alliance partner relationships found for firms in the electronics industry. The electronics industry has transitioned from being dominated by large vertically integrated companies – such as IBM, HP, Toshiba and Fujitsu – into an industry where companies have formed vastly global networks. Further, industries such as electronics are characterized with high market unpredictability, shorter product life cycles, and globalization (Sodhi and Lee, 2007) and that have been relying more on outside suppliers for integration of knowledge and technology (Dedrick et al., 2010). This environment puts greater pressure on firms to make use of the knowledge and technology of their partners to continually produce product

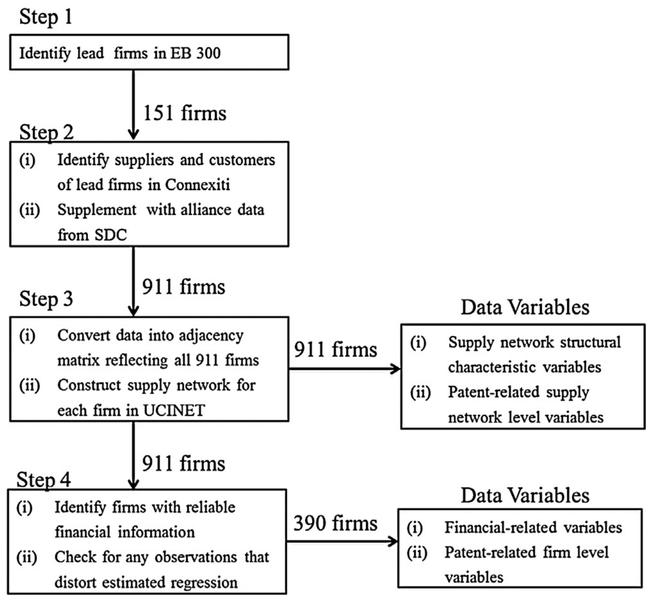

Fig. 4. Steps involved in constructing the dataset for supply network and other data variables.

and process innovations that add customer value. Thus, we find the supply networks of firms in the electronics industry to be a fitting research setting since we are interested in the knowledge flow that arises from the series of supply network relationships. To follow, we summarize our data collection and supply network building approach in four steps. Fig. 4 portrays these steps.

Step 1: First, we identified all firms listed in the EB 300 dataset from 2005 to 2009. The EB 300 dataset is an annual listing of the global 300 electronics firms ranked by revenue (from the sale, service, license or rental of electronics and computer equipment, software, or components), created by Electronics Design, Strategy, and News (EDN). From this list, we retrieved 151 unique firms that were identified as lead companies, contract and original design manufacturers, and component suppliers. We term these firms as "lead firms".

Step 2: Second, we identified the supply network relationships for these 151 lead firms using the Connexiti database. Connexiti is a comprehensive supply chain intelligence database consisting of supplier and customer relationships for nearly 20,000 companies. It contains information on suppliers, customers, and competitors, where information is retrieved from SEC filings, company press releases, website updates, analyst reports, and earning transcripts. Previous research has used Connexiti to study networks (e.g., Basole, 2009). Using each of the 151 lead firms as an initial source, we extracted all supply network relationships in Connexiti that were present during years 2007 and 2008. We then cross-validated and augmented our sample dataset with information from the SDC database, a commonly used source that includes data on strategic alliances as well as supply, manufacturing R&D, marketing and licensing agreements. The SDC database has been used in a number of empirical studies on strategic alliances and inter-firm networks (e.g., Rosenkopf and Padula, 2008; Schilling and Phelps, 2007).

Step 3: Third, by combining the relationship data from step 2 with the initial set of 151 leading firms, our final dataset from which to construct the entire sample of supply networks consisted of 911 firms. Thus, this larger dataset reflecting the full sample of supply networks was used to operationalize the theoretical constructs, supply network accessibility and interconnectedness, which are conjectured in our hypotheses to influence a firm's innovation output.

Step 4: While we had 911 firms from which to build our supply networks and calculate measures related to the structural characteristics, our final sample of firms for the regression analysis step was further reduced due to lack of financial data (e.g., missing data for either creating independent and control variables or for averaging the corresponding measures) or observations identified as outliers. Additionally, finding reliable financial information for private firms was not possible and thus reduced our sample size for analysis to 409 firms. To account for observations whose inclusion unduly distorted the regression model, we calculated the Cook's D values (Cook, 1977) for each observation to find any with very large residuals and with an extreme value on any one of the predictor variables. We excluded any observations that lied above the conventional cut-off of $4/n-k-1$, where n is the sample size and k is the number of predictor variables in the model. This reduced our final sample size for sake of regression analysis to 390 firms.

For our innovation-related variables, we retrieved the patent data from the United States Patent and Trademark Office (USPTO) and Classification and Search Support Information System (CASSIS) Database. We obtained data on patents issued to each company in our sample and cleaned and organized the data on an annual basis. We include a patent in a given year based on its date of application. Using a granted patent's application date allows us to have a closer indication of when the invention occurred, as an invention is estimated to have occurred about three months prior to the patent application date (Darby and Zucker, 2003). Inventions can then be used to trace back a firm's knowledge creation activity. The underlying logic is that inventions serve as a way to instantiate knowledge creation (Schmookler, 1966) and the accumulation of knowledge engrained in inventions is used to facilitate a firm's processes that generate novel actions from a given set of resources (Hargadon and Fanelli, 2002).

3.2. Dependent variable: innovation output

We use the number of patents granted as an indicator of a firm's innovation output (e.g., Penner-Hahn and Shaver, 2005; Rothaermel and Hess, 2007; Shan et al., 1994). Patents serve as a useful measure of innovation output that reflect advancements over existing technology and are externally validated. Hauser et al. (2006) argue that protecting one's lead in technological evolution, and hence achieving competitive advantage, is done by securing patents. Recent studies have also shown that firms that possess a large number of patents are more likely to transform their inventions into a larger number of new products and services introduced to the market (Joshi et al., 2010). Prior research shows that firms in the semiconductor, computer, and communications equipment sectors – all prevalent sectors in our sample – actively patent (Levin et al., 1987). Similar to other studies, we represent the innovation output of a firm by the average number of patent applications granted over three years (2009–2011) (e.g., Benner and Tushman, 2002; Ceccagnoli, 2009). In addition, we have included Table 1 which provides a description of innovation output and of other constructs, their corresponding operationalization, variable type, years for which the data corresponds to, the representative measure, and the data sources.

We acknowledge that there are concerns associated with the use of patent count data that merit discussion. The first concern is the potential right censoring bias when using patent applications granted, as the majority of patent applications are either granted or abandoned within two to three years of application. In fact, over the past 10 years, we verified that the average time from the patent application filing date to the date of disposition (granted or abandoned) has been 2.5 years (USPTO, 2001–2011). In order to mitigate any right censoring bias, our dataset consists of all patents applied for in 2009–2011 that were granted up until January 2014.

Table 1
Description of variables.

Construct	Variable	Variable type	Year	Measure	Data source(s)
Innovation output	Granted patents	Dependent	2009–2011 (Avg)	$Innov_i = Patents_i$	USPTO, CASSIS
Firm size	Natural log of sales	Control	2004–2008 (Avg)	$Firm\_Size_i = Ln(SALES_i)$	Compustat
Firm age	Firm age	Control	N/A	Age_i	Compustat
Industry concentration	Herfindahl–Hirschman index (HHI)	Control	2004–2008 (Avg)	$Ind\_Con_i = \sum_{j=1}^{S} \left(\frac{SALES_i}{SALES_{SIC_j}} \right)$	Compustat
Industry growth	Percent change in sales	Control	2004–2008 (Avg)	$Ind\_SG_i = \%SALES\_Growth_{SIC_j}$	Compustat
Prior innovation	Time-varying patent stock	Control	2006–2008 (Avg)	$Prior\_Innov_i = Time\_Var\_Patent\_Stock_i$	USPTO, CASSIS
Prior knowledge breadth	No. of classes patented in	Control	2004–2008 (Avg)	$Know\_Bre_i = No\_Unique\_Pat\_Classes_i$	USPTO, CASSIS
Supply network accessibility	Information centrality	Independent	2007–2008	$IC_i = \left[\frac{c_{ii} + \left(\sum_{j=1}^{n} c_{jj} - 2\sum_{j=1}^{n} c_{ij} \right)}{n} \right]^{-1}$ $B = D(r) - A + J, \quad C = (c_{ij}) = B^{-1}$	EB300, Connexiti, SDC, UCINET
Supply network interconnectedness	(1 − network efficiency)	Independent	2007–2008	$Interc_i = 1 - \left[\sum_j \left[1 - \sum_q p_{iq} m_{jq} \right] \right] / n_i$	EB300, Connexiti, SDC, UCINET
Absorptive capacity	R&D intensity	Independent	2004–2008 (Avg)	$Abs\_Cap_i = \frac{R\&D_i}{SALES_i}$	Compustat
Supply network partner innovativeness	Average patent stock of a firm's supply network	Independent	2004–2008 (Avg)	$SN\_Part\_Innov_i = \left[\sum_{j=1}^{n_i} Prior\_Patent\_Stock_j \right] / n_i$	USPTO, CASSIS

The second concern is the argument that citation-weighted patent counts better reflect an innovation's quality than patent counts alone. Prior empirical research, however, has established patent count data as reliable in itself by showing the high correlation between patent count and citation-weighted patent measures. In fact, correlations for the two measures were found to be 0.925 ($p < 0.001$) in the electronics and communications industry, 0.973 ($p < 0.001$) in the computers and office machinery industry, and greater than 0.80 ($p < 0.001$) in the semiconductor industry (Hagedoorn and Cloodt, 2003; Stuart, 2000), rendering this assertion more generalizable. Hence, our use of patent counts to reliably proxy the same underlying theoretical construct as citation-weighted patent counts. Further, patent counts have been shown to be positively correlated with new product introductions (Brouwer and Kleinknecht, 1999) and technical capabilities (Hoetker, 2005), and have been regarded as valid and robust indicators of knowledge creation (Trajtenberg, 1987).

3.3. *Independent and moderating variables*

We operationalize two structural characteristics: (1) *information centrality* used to measure supply network accessibility and (2) *network efficiency* to measure the interconnectedness a firm's direct partner supply network. To calculate these two measures, we first construct an undirected binary adjacency matrix reflecting the series of supply network relationships among all firms in our sample. Within our binary adjacency matrix, each cell entry is marked as 1 if there exists a buyer–supplier or alliance relationship between two companies and 0 otherwise. We chose to represent multiple relationships between the same pair of firms as one link in our network for two reasons. First, our primary focus is whether a relationship between two companies exists and not with multiplex relationships. Second, collaborative relationships are typically considered to be bidirectional (Newman et al., 2000). For example, several high-tech products require the integration of sophisticated components that result in ongoing communication and interaction between supply network partners about process and design phases for assembling and testing these products.

We used UCINET 6.365, a social network analysis package (Borgatti et al., 2002), to compute the two independent variable measures. The measures are based on the use of social network analysis, grounded in principles from matrix algebra and graph theory (Wasserman and Faust, 1994). A growing number of supply chain management studies have adopted concepts and tools suggested by Borgatti and Li (2009) that are founded in social network theory (e.g., Autry and Griffis, 2008; Carter et al., 2007; Kim et al., 2011). We describe both measures to follow. For even further clarification on how these measures are calculated, readers can refer to Stephenson and Zelen (1989) and Burt (1992).

3.3.1. *Supply network accessibility*

Supply network accessibility incorporates potential ports of access to knowledge and information across the supply network. We operationalize *supply network accessibility* by using *information centrality* (Stephenson and Zelen, 1989). *Information centrality* is measured by using the harmonic mean length of paths ending at a node i, with this length being smaller if i has many short paths connecting it to other nodes in the network:

$$IC_i = \frac{n}{nc_{ii} + \sum_{j=1}^{n} c_{jj} - 2\sum_{j=1}^{n} c_{ij}} = \left[\frac{c_{ii} + \left(\sum_{j=1}^{n} c_{jj} - 2\sum_{j=1}^{n} c_{ij} \right)}{n} \right]^{-1} \quad (1)$$

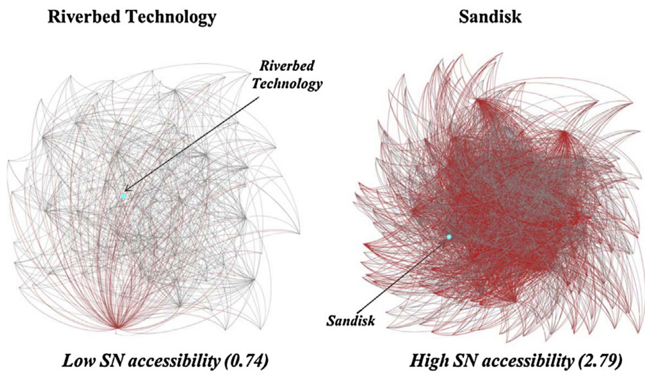

Fig. 5. Comparison of firms with high and low information centrality levels.

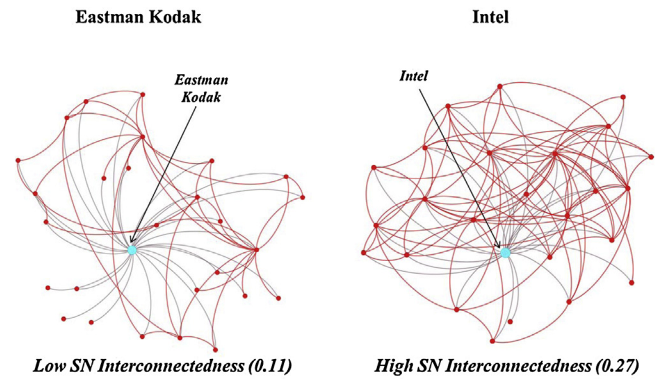

Fig. 6. Comparison of firms with low and high interconnectedness.

where

$$B = D(r) - A + J, \quad C = (c_{ij}) = B^{-1} \tag{2}$$

First, the matrix B is constructed by taking the diagonal matrix $D(r)$ of the number of direct ties firm i has, subtracting it from the adjacency matrix A of the supply network[3], and adding the matrix J with all elements at unity. Next, information centrality scores are calculated using element entries of C, the inverted matrix of B, and the number of firms in the network n. The index has a minimum value of 0, but no maximum value. In our sample, the values for information centrality ranged from a minimum of 0.69 to a maximum of 2.80.

This measure of information centrality focuses on a firm's opportunities to access information and knowledge contained in all paths that originate (and end) at a particular node in a network. This measure is rooted in the theory of statistical estimation, where a path connecting two nodes is considered as a signal and the noise in the transmission of the signal is measured by the variance of this signal. The measure of information available through each transmission would then be the reciprocal of the variance (Stephenson and Zelen, 1989). For the sake of our study, information centrality serves as a measure of *supply network accessibility* to represent the speed and extent of opportunities a firm has to access information and knowledge from other members in the supply network.

Fig. 5 illustrates a comparison of two companies with different levels of accessibility within the overall supply network. The darker lines reflect the connections shared among that focal firm's partners, and the focal firm is indicated by the biggest node in the graph. Here we highlight the wider supply network that a focal firm has access to via its indirect connections. The firm Riverbed Technology appears to have a relatively low level of accessibility to other members in the supply network compared with another firm, Sandisk, which appears to possess a relatively high level of supply network accessibility.

3.3.2. *Supply network interconnectedness*

We capture *supply network interconnectedness* by assessing the number of shared relationships that exist between the supply network partners of a focal firm. As mentioned earlier, we are also interested in capturing the extent to which a firm's supply network partners are densely (sparsely) connected. Assessing shared relationships helps provide insights into how closely knit a focal firm's partners are with each other and into possible redundant ties that are built into the supply network. More formally, *network efficiency* accounts for the level of *supply network interconnectedness* by adapting the efficiency equation from Burt (1992):

$$Interc_i = 1 - Effic_i = \frac{1 - \left[\sum_j \left[1 - \sum_q p_{iq} m_{jq}\right]\right]}{n_i} \tag{3}$$

where p_{iq} is the proportion of focal firm i's ties invested in the relationship with q, m_{jq} is the marginal strength of the tie between members j and q (that are both directly connected to i) and n_i is the total number of direct partners of focal firm i. Since our supply network representations are binary, the values of m_{jq} are set to 1 if a tie is present between members j and q and 0 otherwise.

The notion of network efficiency suggests that, if a focal node has at least one pair of direct sources who are also directly connected to each other, then its network is considered to be inefficiently connected. Thus, a network is considered to be inefficiently connected in a sense that there is at least one tie in the network that indirectly connects the focal node to the same source of knowledge, resource, or information. This tie would be considered as a redundant tie. Therefore, we use network efficiency to measure how interconnected a supply network is based on the number of redundant ties present in the supply network. Based on this operationalization, a larger network efficiency score would correspond to a lower level of *supply network interconnectedness* and vice versa.

The values for the measure of network efficiency range from 0 to 1, where a value of 1 indicates that all of a firm's partners share no ties to each other. Thus, since this measure works in the opposite direction as our hypothesized construct and spans from 0 to 1, we calculated *supply network interconnectedness* as (1 − *network efficiency*) to help avoid ambiguity in interpreting its hypothesized effect. In our sample, the values for network efficiency ranged from a minimum of 0.33 to a maximum of 1. This reverse coding resulted in minimum and maximum *supply network interconnectedness* values of 0 and 0.67, respectively. The illustrations in Fig. 6 provide a comparison of two companies with different supply network structures, but with a similar number of direct ties. The lighter lines indicate a focal firm's direct connections, while the darker lines reflect the connections among that focal firm's partners. The focal firm is indicated by the biggest node in the graph. Eastman Kodak appears to have a supply network characterized by low interconnectedness and hence a high level of efficiency. Conversely, Intel appears to have a highly interconnected supply network with the multitude of ties offering several ports of access to knowledge, resources, and information. Network efficiency has been used in other interfirm studies as a redundancy based measure of structural holes at the local level, based only on ties between a focal firm's direct partners (Ahuja, 2000; Paruchuri, 2010; Schilling and Phelps, 2007).

[3] In the adjacency matrix A, each element a_{ij} is equal to 1 if a tie exists a tie between firm i and firm j, and 0 otherwise.

3.3.3. Absorptive capacity: R&D intensity

A firm's absorptive capacity – the ability to recognize, assimilate, and deploy outside knowledge – is a crucial element affecting its innovation output levels. Research has suggested that a firm's absorptive capacity is largely a function of its investment in R&D and its level of prior related knowledge (Cohen and Levinthal, 1990). The rationale is that firms conducting their own R&D are better equipped to make use of externally available information. While the notion of absorptive capacity can also be explained in terms of a firm's direct involvement in the manufacturing process or the cognitive structures that underlie learning, we rely on the more readily available measure of investment in R&D as a proxy for a firm's ability to recognize and exploit new knowledge from external sources in their supply network (Cohen and Levinthal, 1990). Thus, we included *R&D intensity* as it can contribute to a firm's ability to absorb outside knowledge (Rothaermel and Hess, 2007). We calculated *R&D intensity* as the R&D expenditures measured as percentage of total sales.

3.3.4. Supply network partner innovativeness: average patent stock of firm's partners

To capture *supply network partner innovativeness*, we consider the total knowledge accumulated from each supply network partner, based on the partner's respective history of patenting activities. Assessing the amount of knowledge accumulated by each of its partners over time gives the firm a better sense of the magnitude of knowledge that may spill over as the relationship develops. We operationalize *supply network partner innovativeness* as follows. First, we calculated the average patent stock of each supply network partner by summing a partner's patent stock over the previous five years (2004–2008). Next, for each firm, we then measured supply network partner innovativeness by taking the mean of the average patent stock of all of a firm's partners. Lastly, we normalized this measure of supply network partner innovativeness for use in the analysis (Narin et al., 1987). Previous scholars point out how a firm's current technological stance is often dependent on its previous level of technological know-how, due to the cumulative nature of technology. A firm's level of *patent stock*, also referred to as technical capital in prior literature, can be seen to represent the depth of a firm's technological resources (Silverman, 1999). *Patent stock* has been used in previous studies looking at high-tech industries to assess technological impact, calculated as a firm's patenting activity in the previous five years (e.g., Ahuja, 2000; Vanhaverbeke et al., 2009). Thus, we calculated the cumulative patent stock of each firm in the sample from 2004 to 2008.[4]

3.4. Control variables

As shown in Table 1, we control for the following variables: Firm size, firm age, prior innovation, knowledge breadth, lead firm indicator, and industry-level control variables (industry concentration and industry growth). Financial data for the moderating and control variables was retrieved from the Compustat database and cross-examined using the Mergent Online database. *Firm size* may influence a firm's level of innovation output, as larger firms have more financial means and greater resources to invest in innovation-related activities than smaller firms. Interestingly, firm size can both positively or negatively influence its innovation output (Teece, 1992). We controlled for firm size using the natural log of sales. Next, we controlled for *firm age*, as older firms are expected to leverage more of their existing technological competencies while younger firms are expected to experiment more with new technologies (Sorensen and Stuart, 2000). *Firm age* was calculated as the number of years from the date of the firm's founding to the current year of 2013. We also include the following two industry-level control variables, with industry defined at the two-digit Standard Industry Classification (SIC) level: industry concentration (captured by the Herfindahl–Hirschman Index (HHI)) and industry growth (captured by the percentage change in annual industry sales). The use of industry controls help ensure that our findings are robust to industry effects, as industry may influence a firm's patenting behavior. Next, we incorporate two variables into our model to control for past innovation activity: prior innovation and knowledge breadth. Prior innovation is used to capture the time-varying patent stock of a firm, while knowledge breadth is used to capture the number of unique classes a firm has patented in. Lastly, we control for lead firms in our sample, to isolate the effect of such firms on innovation output. We operationalize this as a binary variable with a "1" indicating that the focal firm was one of the lead firms and a "0" otherwise.

3.5. Model specification

We operationalize innovation output by using the number of granted patents as our dependent variable. A count variable that takes on only non-negative integer values makes a linear regression model inappropriate as it assumes the distribution of residuals to be homoscedastic, normally distributed. This could lead to coefficient estimates that are both biased and inconsistent (Greene, 2003). Poisson and negative binomial regression are more appropriate models for count data. Because of the presence of overdispersion in our patent data, the assumption of Poisson regression that the mean and variance are equal does not hold. The negative binomial model accounts for overdispersion and helps avoid spuriously high levels of significance due to coefficients whose standard errors are underestimated (Cameron and Trivedi, 1986). By inspecting the likelihood ratio test, we found strong evidence for the negative binomial model as more appropriate than the Poisson model for our data ($p < 0.001$).

The negative binomial model has the following form (Hilbe, 2011):

$$\mathscr{L} = \sum_{i=1}^{n} \left\{ y_i \ln\left(\frac{\alpha \exp(x_i'\beta)}{1+\alpha \exp(x_i'\beta)}\right) - \frac{1}{\alpha}\ln(1+\alpha\exp(x_i'\beta)) + \ln\Gamma\left(y_i + \frac{1}{\alpha}\right) - \ln\Gamma(y_i+1) - \ln\Gamma\left(\frac{1}{\alpha}\right) \right\} \quad (4)$$

The above equations for the model are expressed as log-likelihood functions, as is typical for a count model. In the above equations, y_i refers to the outcome variable measured by patent count, the x_is refer to each explanatory variable (*firm size, firm age, industry concentration, industry growth, prior innovation, knowledge breadth, lead firm, supply network accessibility, supply network interconnectedness, absorptive capacity*, *supply network partner innovativeness, supply network accessibility * supply network interconnectedness, supply network accessibility * absorptive capacity, supply network interconnectedness * supply network partner innovativeness*), α reflects the value of the heterogeneity or overdispersion parameter, and β represents the model coefficients.

4. Results

We ran all analyses in STATA Version 13. We first report summary statistics by industries defined at the two-digit SIC level in Table 2. The descriptive statistics and simple correlations are presented in Table 3. We also took several measures to account for

[4] As a note, we also ran our analysis using the previous three and four years separately. Our results were robust to these changes.

Table 2
Summary statistics by industries defined at the two-digit Standard Industry Classification (SIC) level.

SIC	Industry	# observations	Firm innovation	Firm size	Firm age	Industry concentration	Industry growth	Prior innovation	Prior knowledge breadth	SN accessibility	SN interconnectedness	Absorptive capacity	SN partner innovativeness
2800–2899	Chemicals & Allied Products	4	17.3	5.44	37	0.25	0.097	−0.12	36	2.15	0.097	0.037	0.075
3000–3099	Rubber & Misc. Plastics Products	3	2.67	6.44	17.3	0.19	−0.0090	0.038	12	1.41	0.019	0.023	0.025
3300–3399	Primary Metal Industries	4	10.3	7.22	19	0.31	0.056	0.29	4.50	1.90	0.25	0.012	0.025
3500–3599	Industrial Machinery & Computer Equip.	79	12.1	5.73	20.8	0.41	0.11	−0.092	15.3	2.02	0.19	0.15	0.079
3600–3699	Electronic, Electrical Equip. & Compts., Not Computer Equip.	185	27.4	5.71	22.3	0.19	0.074	0.017	18.8	2.18	0.18	0.18	0.080
3700–3799	Transportation Equip.	3	13	8.87	62	0.34	0.083	−0.25	80	1.87	0.27	0.033	0.049
3800–3899	Instruments & Related Products	28	11.9	5.38	22.7	0.27	0.066	−0.081	15.7	2.01	0.17	0.16	0.064
3900–3999	Misc. Manufacturing Industries	2	8	8.12	20.5	0.33	0.18	−0.076	15	2.57	0.028	0.029	0.080
4800–4899	Communications	10	30.8	6.58	20.7	0.16	0.044	0.011	12.5	1.41	0.083	0.073	0.018
5000–5099	Wholesale Trade – Durable Goods	9	2.22	7.80	32.4	0.34	0.075	0.15	2.67	1.71	0.068	0.00088	0.028
7300–7399	Business Services	63	37.7	5.72	19.2	0.20	0.063	0.022	12.8	1.81	0.11	0.17	0.056
Total	All Industries	390	23.7	5.82	22.1	0.25	0.078	−0.0091	16.8	2.04	0.17	0.16	0.071

multicollinearity. First, we used the grand mean-centered values of all explanatory variables used in the interaction terms, mitigating multicollinearity. Second, we ensured that the variance inflation factor (VIF) scores for each predictor variable were below a value of 10, indicating that multicollinearity is not an issue in the given dataset (Neter et al., 1996). Each of the VIF scores for our dataset met this requirement (mean score of 1.43) after we mean-centered the necessary variables.

4.1. Main results

The results of the negative binomial regression are presented in Table 4. The effects are introduced sequentially in models 1–5 to help ensure model stability and to make sure that any significant effect is robust to the inclusion of other effects. For each model, we performed Wald tests based on the null hypothesis that that all of the estimated coefficients that were not present in the previous model are equal to zero. The χ^2 statistics and significance levels are found below the log likelihood values in Table 4. Model 1 includes only the control variables. Some of the control variables are significant. Specifically, firm size, and prior knowledge breadth are shown to positively affect the level of innovation output. Conversely, firm age and industry concentration have a negative influence on innovation output, suggesting that older firms tend to rely on more heavily on existing technology and less on patenting new innovations and that more concentrated industries have lesser innovation output. Model 2 includes the main effects of supply network accessibility, supply network interconnectedness, absorptive capacity, and supply network partner innovativeness. The results suggest that the level of innovation output increased with an increase in supply network accessibility ($p < 0.001$), thus providing support for Hypothesis 1. The model displays an insignificant relationship between supply network interconnectedness and innovation output, showing lack of support for Hypothesis 2. Models 3 and 4 incrementally include interactions related to Hypotheses 3 and 4. Lastly, Model 5 includes the interaction related to Hypothesis 5 and represents the full model.

As seen in Table 4, the positive association between supply network accessibility and innovation output (Hypothesis 1) remains throughout to the full model. Also, the coefficients reflecting the effect of supply network interconnectedness are all positive as expected (Hypothesis 2) albeit only significant in the full model. Further, Hypothesis 3 that posited supply network interconnectedness has a positive and significant interaction effect on supply network accessibility is at least partially supported ($p < 0.10$). Moreover, the results suggest that the positive association between supply network accessibility and innovation output is positively moderated by a firm's absorptive capacity ($p < 0.05$), thus providing support for Hypothesis 4. Lastly, the positive and significant association between the moderation of supply network partner innovativeness on supply network interconnectedness ($p < 0.05$) indicates support for Hypothesis 5. The results shown in Model 5 are an improvement over all previous models (e.g., compared with Model 4, likelihood ratio test statistic: 4.92; $p < 0.05$).

4.2. Moderating effects

Further interpretation of each moderating effect can be enriched by using interaction plots of the variables of interest. We graph the predicted innovation output with changes in each corresponding variable, using high and low values of the variable values as one standard deviation above and below the mean, respectively. Fig. 7 shows the plot of the interaction between a firm's supply network accessibility and its level of interconnectedness. The "Low SN interconnect." line relates to the moderating effect of supply network interconnectedness, and depicts the slope of the effect of

Table 3
Descriptive statistics and correlations.[a]

	Variable	Mean	Std. Dev.	1	2	3	4	5	6	7	8	9	10	11	12
1	Firm innovation	23.65	82.84	1.00											
2	Firm size	5.82	2.20	0.48	1.00										
3	Firm age	22.11	12.05	0.18	0.37	1.00									
4	Industry concentration	0.25	0.17	−0.08	0.00	0.08	1.00								
5	Industry growth	0.08	0.05	−0.03	0.04	−0.02	0.19	1.00							
6	Prior innovation	−0.01	0.35	−0.16	−0.18	−0.10	−0.05	−0.02	1.00						
7	Prior knowledge breadth	16.78	27.53	0.66	0.65	0.32	−0.04	0.00	−0.27	1.00					
8	Lead firm	0.22	0.42	0.41	0.67	0.29	0.03	0.09	−0.18	0.60	1.00				
9	SN accessibility	2.04	0.61	0.16	0.32	0.13	−0.07	0.13	−0.02	0.30	0.45	1.00			
10	SN interconnectedness	0.17	0.13	0.07	0.15	0.04	−0.01	−0.01	−0.09	0.11	0.10	0.30	1.00		
11	Absorptive capacity	0.16	0.20	−0.08	−0.38	−0.20	−0.10	0.13	−0.05	−0.11	−0.15	−0.07	−0.03	1.00	
12	SN partner innovativeness	0.07	0.12	0.07	0.23	0.11	−0.01	0.14	0.02	0.20	0.35	0.53	0.00	−0.01	1.00

[a] $N = 390$ observations. All correlations with magnitude $>|0.095|$ are significant at $p < 0.05$ level.

supply network accessibility on patents when the value of supply network interconnectedness is set to one standard deviation below its (mean-centered) mean. In contrast, the "High SN interconnect." line reflects the slope of the effect of supply network accessibility on innovation output when the value of supply network interconnectedness is set to one standard deviation above its (mean-centered) mean. High levels of supply network interconnectedness are shown to positively reinforce the effect of supply network accessibility on the firm's innovation output.

Fig. 8 shows the plot representing the moderating effect of absorptive capacity on a firm's supply network accessibility. The "Low Abs. Cap." line relates to the moderating effect of absorptive capacity, and depicts the slope of this effect on supply network accessibility on innovation output when the value of absorptive capacity is set to one standard deviation below its (mean-centered) mean. In contrast, the "High Abs. Cap." line reflects the slope of the effect of supply network accessibility on innovation output when the value of absorptive capacity is set to one standard deviation above its (mean-centered) mean. High levels of absorptive capacity are shown to positively reinforce the relationship between supply network accessibility and innovation output.

Lastly, Fig. 9 shows the plot representing the moderating effect of supply network partner innovativeness on supply network interconnectedness. The "Low SN Partner Innov." line relates to the moderating effect of supply network partner innovativeness, and depicts the slope of this effect on supply network

Table 4
Negative binomial regression model – innovation output (2009–2011 patents).[a]

Variables	Model 1	Model 2	Model 3	Model 4	Model 5
Controls					
Firm size	0.62***	0.67***	0.66***	0.66***	0.68***
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)
Firm age	−0.01*	−0.01	−0.01*	−0.01	−0.01
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Industry concentration	−1.90***	−1.69***	−1.47***	−1.41***	−1.39***
	(0.39)	(0.38)	(0.39)	(0.39)	(0.39)
Industry growth	1.94	1.56	1.23	1.85	2.06
	(1.43)	(1.39)	(1.37)	(1.38)	(1.37)
Prior innovation	0.24	0.21	0.16	0.18	0.18
	(0.18)	(0.18)	(0.17)	(0.17)	(0.17)
Prior knowledge breadth	0.02***	0.02***	0.02***	0.02***	0.02***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Lead firm	0.09	−0.24	−0.33	−0.22	−0.32
	(0.19)	(0.21)	(0.21)	(0.21)	(0.21)
Direct effects					
SN accessibility[b]		0.50***	0.68***	0.69***	0.63***
		(0.15)	(0.15)	(0.15)	(0.15)
SN interconnectedness[b]		0.11	0.87	0.83	1.28*
		(0.54)	(0.57)	(0.56)	(0.59)
Absorptive capacity[b]		0.93*	0.80	1.24*	1.30*
		(0.46)	(0.46)	(0.54)	(0.53)
SN partner innovativeness[b]		−0.36	−0.30	−0.52	0.12
		(0.64)	(0.61)	(0.61)	(0.69)
Moderation effects					
SN accessibility * SN interconnectedness			3.14***	3.19***	2.00
			(0.93)	(0.93)	(1.07)
SN accessibility * absorptive capacity				1.80*	1.78*
				(0.87)	(0.85)
SN interconnectedness * SN partner innovativeness					19.58*
					(8.79)
Constant	−1.97***	−2.28***	−2.30***	−2.41***	−2.46***
	(0.28)	(0.30)	(0.30)	(0.30)	(0.30)
Log likelihood	−1035.64	−1025.42	−1019.91	−1016.45	−1013.99
LR test		20.44***	11.02***	6.91**	4.92*
N	390	390	390	390	390

[a] Standard errors in parentheses: ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$.
[b] Variables were grand mean-centered.

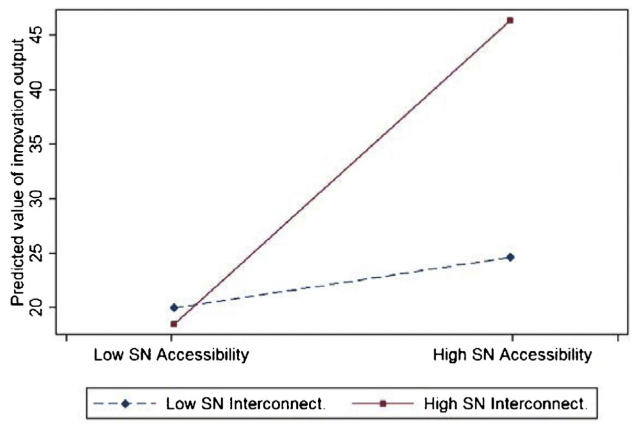

Fig. 7. Moderating effect of supply network interconnectedness on supply network accessibility.

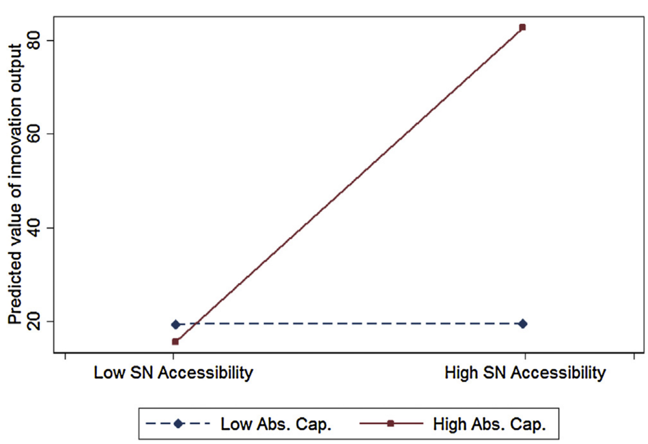

Fig. 8. Moderating effect of absorptive capacity on supply network partner innovativeness.

interconnectedness on innovation output when the value of supply network partner innovativeness is set to one standard deviation below its (mean-centered) mean. In contrast, the "High SN Partner Innov." line reflects the slope of the effect of supply network interconnectedness on innovation output when the value of supply network partner innovativeness is set to one standard deviation above its (mean-centered) mean. High levels of supply network partner innovativeness are shown to positively reinforce

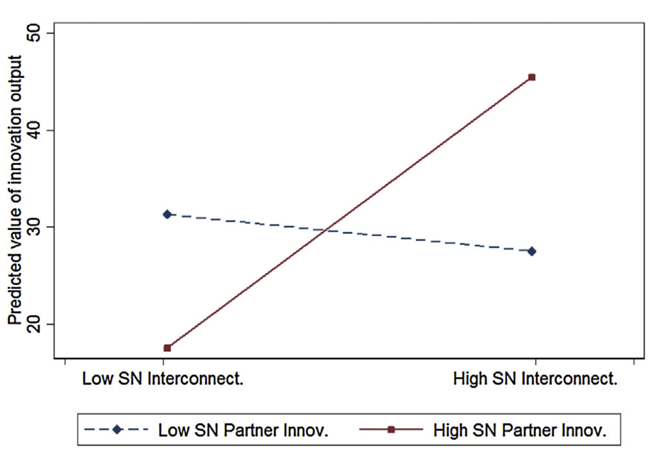

Fig. 9. Moderating effect of supply network partner innovativeness on supply network interconnectedness.

the relationship between supply network interconnectedness and innovation output.

4.3. Robustness checks

We also estimated the model using a zero-inflated negative binomial. This alternative specification helps to account for the large number of zeros that are present in our dataset (27.18% of firms). While many firms decide to protect their innovative ideas through patents, others may wish to protect theirs through other means, such as trade secrets. However, this decision is unobserved. This leads to two types of zeros: those zeros due to a firm not reporting their innovation via a patent and zeros due to the firm not actually having any innovation to patent for a given year. One way to account for this is to use the zero-inflated negative binomial approach, first introduced by Lambert (1992). This model first generates two separate models: a negative binomial count model and the logit model for predicting excess zeros (Hilbe, 2011). First, the logit model is generated to capture the zeros for those firms that may had decided to report their innovation by means of a patent ("certain zeros"), predicting the probability of a firm falling in this category or not. Second, the negative binomial model is generated to predict the counts for those firms that may have decided not to report their innovation by patenting it (not a part of the "certain zeros"). In the next step, the negative binomial and logit models are combined, forming the following zero-inflated negative binomial–logit model, (referred to as the zero-inflated negative binomial model for short). Results of this alternative model can be seen on Table 5. To test for the suitability of the zero-inflated negative binomial over the negative binomial model, we inspect the commonly-used Vuong test (Vuong, 1989). We followed the standard that the zero-inflated model is suitable if the Voung statistic is greater than 1.96 (Long, 1997). The Vuong test yielded values above the threshold for Models 1–5 (e.g. 2.96 in Model 5), signifying that the zero-inflated variant as an appropriate alternative model. We obtained results structurally similar to our negative binomial model. The hypothesized effects are in the same direction as our negative binomial model. Compared to the negative binomial regression results, we find support for Hypotheses 1, 3, 4 and 5, whose hypothesized effects are all significant at $p < 0.05$.

Lastly, we had not proposed a quadratic functional form that could possibly describe the relationship between accessibility and innovation output (Hypothesis 1). However, to verify possible non-linearity, we provide an alternative model with the squared term of accessibility as a control variable in the regression models. Similarly, though we do not hypothesize for the curvilinear effect for interconnectedness (Hypothesis 2), we tested an alternative model to control for its possible non-linearity. Thus, in a separate regression, we have tested an alternative model using both squared terms of accessibility and interconnectedness as controls. Both the squared terms are not significant and their inclusion leads to similar results that agree with the findings in our model without the terms.

4.3.1. Endogeneity

We also took further measures to account for potential issues of endogeneity arising in our model. There is a possibility that firms exhibiting high innovation output may select or come to occupy favorable structural positions and influence its partners to become more inter-linked by being able to form network ties that lead them to display such structural characteristics. Thus, if this source of endogeneity existed, the error terms of the endogenous explanatory variables would be correlated with the error terms of the dependent variable, leading to biased and inconsistent results (Greene, 2003).

Table 5
Alternative zero-inflated negative binomial model – innovation output (2009–2011 patents).[a]

Variables:	Model 1	Model 2	Model 3	Model 4	Model 5
Controls					
Firm size	0.62***	0.66***	0.65***	0.65***	0.66***
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)
Firm age	−0.01*	−0.01*	−0.01*	−0.01*	−0.01*
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Industry concentration	−2.05***	−1.91***	−1.68***	−1.61***	−1.60***
	(0.36)	(0.36)	(0.36)	(0.36)	(0.36)
Industry growth	1.48	1.32	0.96	1.62	1.80
	(1.35)	(1.32)	(1.30)	(1.31)	(1.30)
Prior innovation	0.21	0.19	0.14	0.15	0.16
	(0.17)	(0.17)	(0.16)	(0.16)	(0.16)
Prior knowledge breadth	0.02***	0.01***	0.01***	0.01***	0.01***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Lead firm	0.19	−0.08	−0.17	−0.05	−0.15
	(0.17)	(0.19)	(0.19)	(0.19)	(0.20)
Direct effects					
SN accessibility[b]		0.47***	0.64***	0.65***	0.60***
		(0.14)	(0.14)	(0.15)	(0.15)
SN interconnectedness[b]		−0.26	0.49	0.45	0.85
		(0.51)	(0.54)	(0.53)	(0.55)
Absorptive capacity[b]		0.67	0.51	0.85	0.91
		(0.45)	(0.44)	(0.56)	(0.55)
SN partner innovativeness[b]		−0.58	−0.49	−0.72	−0.16
		(0.57)	(0.54)	(0.55)	(0.62)
Moderation effects					
SN accessibility * SN interconnectedness			3.25***	3.34***	2.19*
			(0.88)	(0.88)	(1.01)
SN accessibility * absorptive capacity				2.10*	2.03*
				(0.93)	(0.91)
SN interconnectedness * SN partner innovativeness					18.04*
					(8.04)
Constant	−1.70***	−1.98***	−2.00***	−2.10***	−2.15***
	(0.28)	(0.29)	(0.29)	(0.30)	(0.30)
Log likelihood	−1005.38	−997.9	−991.39	−987.39	−984.87
LR test		20.44***	11.02***	6.91**	4.92*
Voung	2.98**	2.98**	2.82**	2.91**	2.96**
N	390	390	390	390	390

[a] Standard errors in parentheses: *** $p<0.001$, ** $p<0.01$, * $p<0.05$.
[b] Variables were grand mean-centered.

To address the potential endogeneity problem between each supply network structural characteristic and innovation output, a two-stage least squares (2SLS) estimation procedure was adopted. In the first stage, each structural characteristic was regressed on all assumed exogenous variables on two separate regressions – one for supply network accessibility and another for supply network interconnectedness – in order to obtain predicted values for these potentially endogenous variables. In the second stage, the predicted values from the first stage were included as independent variables to replace the values of the assumed endogenous variables.

Before the 2SLS was executed, we had to identify instrumental variable candidates that met validity requirements. First, in a regression with only assumed exogenous variables from the original count model, we identified candidates that were not significantly correlated with innovation output at the 5% significance level. From this step, we chose industry sales growth and prior innovation as instruments for both structural characteristics (significance levels can be seen in Tables 4 and 5; we also verified this using a joint χ^2 test). We also used prior innovation as an instrument as a primary concern here is whether high patenting activity significantly influences network structure. Second, we identified two other variables related to structural characteristics that were not significantly correlated with innovation output but significantly correlated with at least one of the assumed endogenous structural characteristics: degree centrality and number of pairs. Degree centrality is a more simplified measure of a firm's structural position found in network analysis literature, and is measured as the number of supply network partners (i.e., direct ties) they have. The number of pairs is a measure capturing the total number of unordered pairs of distinct nodes that are directly connected to a focal node (Wasserman and Faust, 1994). In our case, this measures the total number of potential ties between every partner pair, of all partners who share a direct tie with the focal firm. To sum, we chose industry sales growth, prior innovation, degree centrality, and number of pairs as instruments for potentially endogenous variables supply network accessibility and supply network interconnectedness. The table in Appendix A (Models (1) and (2)) shows the results of the first stage regressions for supply network accessibility and supply network interconnectedness.

Though we cannot directly test the statistical independence of an instrument from the error terms of the dependent variable, we can assess the adequacy of our instruments using a test of overidentifying restrictions (Baum, 2006). A test of the overidentifying restrictions also suggests that the chosen instruments are exogenous. Specifically, after regressing on innovation output with exogenous variables from the original count model plus all instruments, the null hypothesis that the instruments are uncorrelated with the error term of the dependent variable (innovation output) cannot be rejected at 5% significance level. We examined the explanatory power of instruments when used as independent variables in regressions on each assumed endogenous variable. Joint tests of the null hypothesis that the corresponding instruments have no effect on supply network accessibility and supply network interconnectedness show χ^2 statistics of 59.34 ($p<0.001$) and 5.60 ($p<0.001$), respectively. Thus, we can reject the null that the corresponding instruments have no effect, suggesting that there is a significant correlation with the instruments and network structural characteristics. Taking stock of all validity checks, we conclude that we have valid instruments for each potentially endogenous variable.

Appendix A (Model (3)) shows the results of the second stage regression with the predicted values from the first stage included as independent variables to replace the values of the assumed endogenous variables, supply network accessibility and supply network interconnectedness. After running the 2SLS, we performed a Durbin–Wu–Hausman postestimation test of endogeneity, which adds the error terms from the first stage (using robust variance estimates) and separately tests whether they are correlated with error terms in the original count model (Cameron and Trivedi, 2009). Using the error terms from the first stage for the assumed endogenous variables in separate tests, both endogeneity test statistics had p-values greater than 0.10, indicating that we fail to reject the null that these variables are exogenous. In other words, the endogeneity tests associated with supply network accessibility and supply network interconnectedness were both insignificant. Hence, the parameter estimates for these variables in our original count model do not appear to be unduly influenced by endogeneity.

5. Discussion

Scholars in operations management have exhorted future research to examine the structural dimension of social networks, specifically, accounting for the embedded nature of buyer-supplier dyads (Autry and Griffis, 2008; Villena et al., 2011). Building on extant research, this study examines the association between the structural characteristics (supply network accessibility and supply network interconnectedness) of a firm's supply network and its innovation output. We find that supply network accessibility is positively associated with innovation output. Conversely, no significant association exists between supply network interconnectedness and innovation output. However, there is at least a partially significant interaction effect between these structural characteristics and innovation output. Besides investigating the effects of structural characteristics, we also examine the moderating role of absorptive capacity and supply network partner innovativeness in strengthening the associations between the structural characteristics and innovation output. The findings suggest significant moderating roles of both absorptive capacity and supply network partner innovativeness. Table 6 summarizes our overall findings.

5.1. Theoretical implications

Our study contributes to the literature by addressing the interface between supply networks and innovation by investigating how a firm can accrue knowledge and information flow benefits from its supply network to enhance its innovation output. Integration and collaboration with supply network partners has been recognized not only to improve product quality, service levels, and revenue enhancements, but also as a key source of innovation. We extend this literature by explicitly examining supply networks and their two key inherent structural characteristics as a source of innovation.

The results for Hypothesis 1 illustrate that the level of network accessibility that a firm has to the resources and knowledge assets of its supply network partners – as derived from its structural position in the supply network – influences its innovation output. This finding is in agreement with the evidence in the literature showing that firms with high network accessibility experience a greater volume and diversity of information from their partners in the supply network in which they operate (Schilling and Phelps, 2007). Findings for the main effect of supply network interconnectedness on innovation output are inconclusive, indicating lack of support for Hypothesis 2. Our main premise was that this interconnectedness helps foster collaborative initiatives that provide access to knowledge, resources, and information from other partners in the supply network.

Lack of support for a direct relationship between supply network interconnectedness and innovation output suggests that interconnectedness, in isolation, may not be a significant driver of a firm's innovation output. One explanation may be that, while certain levels of supply network interconnectedness benefit firms in terms of operating performance, there are other contingencies in the context of innovation that make its effect on knowledge creation and innovation outcomes unclear (Vanhaverbeke et al., 2009). In this regard, the partial support for Hypothesis 3 provides some evidence that while the main effect of supply network interconnectedness is not significant, it may help moderate the influence of supply network accessibility on innovation output. In other words, we find some indication that higher levels of network interconnectedness may strengthen the beneficial effect of supply network accessibility on innovation output. This evidence suggests that establishing highly interconnected supply networks may facilitate collaboration among supply network partners and act as one of the moderating mechanisms to magnify the positive effect of supply network accessibility on innovation output.

Our study also provides evidence that while structural characteristics in a supply network can enable information and knowledge flows to enhance innovation output, this association can be capitalized by two knowledge variables, absorptive capacity and supply network partner innovativeness. The results show that investing more in R&D, as a manifestation of absorptive capacity, can be used to positively moderate the effects of supply network accessibility on innovation output. An example echoing this phenomenon is Milliken & Company, an innovative firm possessing expertise in several areas including specialty chemical, floor covering, and performance materials. Not only do they invest heavily in R&D themselves but also have access to a network of suppliers and partners (PR Newswire, 2013). As of 2013, Milliken & Company had developed one of the largest collections of patents held by a private U.S. company.

Finally, the finding that supply network partner innovativeness moderates the relationship between supply network interconnectedness and innovation output highlights the advantage of firms in drawing from the knowledge assets of their supply network partners to further their innovation output. In other words, while operating in highly interconnected supply networks has potential for firms to facilitate knowledge flow and sharing opportunities among partners, it is the magnitude of knowledge availability among a firm's partners that actually enhance any trickle down effects that improve the firm's subsequent efforts to generate innovation output (Stuart, 2000).

5.2. Managerial implications

This study provides suggestions that can have managerial implications related to the influence of a firm's supply network structure on its innovation output. First, managers should recognize the important role of structural capital – in the form of supply network structural characteristics – in maintaining and facilitating knowledge creation. This perspective would require firms to consider the value of their key supply network partners that exists directly and indirectly through each partner's extended network of relationships. If firms want to accrue benefits in their innovation activity from their supply network, they should proactively cultivate the structural characteristics of their supply networks that lead to greater innovation output because of superior knowledge-sharing practices among their suppliers (Von Hippel, 1988). Accordingly, managers might need to promote more interactions within their supply network to help facilitate effective information and

Table 6
Summary of findings.

Hypothesis	Factor(s)	Impact on innovation	Support?
1	SN accessibility	Positive	Yes
2	SN interconnectedness	Positive	No
3	SN accessibility * SN interconnectedness	Positive	Partially supported
4	SN accessibility * absorptive capacity	Positive	Yes
5	SN interconnectedness * SN partner innovativeness	Positive	Yes

knowledge flow throughout the supply network. Managing structural characteristics will also help to focus their strategy on maximizing opportunities to access external knowledge (Nyaga et al., 2010) as well as solicit supply network partner input and feedback during its innovation processes (Koellinger, 2008). In other words, adopting a supply network perspective can help firms to focus on knowledge and information flow opportunities residing in its supply network to benefit its innovation output.

Second, the perspective of supply network partner value can also apply to the manager's future selection and supply network configuration strategy. Autry and Griffis (2008) note the prospect of future research to investigate a firm's decision "to invest in competitive intelligence that can be used to optimize the structure of the supply chain by identifying the most attractive partnering opportunities" (p. 168). This sort of strategic approach suggests that a firm can focus more on investments that promote structural changes (related to the supply network structure) or relational ones (related to direct investments in relationships with partners in its supply network) that facilitate more effective knowledge and information sharing. The former approach could also result in more direct intervention of a buying firm to reconfigure its suppliers' external networks or communications structures (Choi and Kim, 2008). For supply chain managers whose focus of building competitive advantage is through innovation leadership, our study suggests significant benefits from assessing a firm's level of supply network accessibility, and the level of interconnectedness among supply network partners, on influencing the lead time of information and knowledge flow and increasing ports of access for knowledge and information sharing.

Finally, the findings also suggest that firms should not overlook the innovativeness of their partners since working with innovative partners may have a direct bearing on their own innovation capabilities. This finding is in line with the trend of firms building core competencies in-house but outsourcing non-core competencies, making them more dependent on the knowledge and expertise of their supply network partners to avoid sub-optimal solutions to problems, to innovate, and to adapt (Zacharia et al., 2011). For example, Mercedes-Benz recently announced that to maintain its high innovative activity it will "rely on its internal expertise and its network of suppliers around the world" (Reuters, 2013).

5.3. Limitations and directions for future research

While the insights found in our study are important, we acknowledge that our study has limitations. Part of the motivation for this study was to help address the call for deeper structural analysis accounting for the embedded nature of buyer-supplier dyads (Autry and Griffis, 2008; Villena et al., 2011). Thus, we drew our analysis from a unique dataset that allowed us to use a supply network lens and incorporate the embedded nature of each supply network partner dyad. Our study thus emphasizes the embeddedness of a firm and the interconnectedness of its partners. We did not incorporate relationship strength, reflecting the importance or value of each supply network link. Some suppliers may be providing very standard components with little value added whereas other suppliers may be providing a critical component that adds considerable market value. Also, firms may rely heavier on a certain customer based on the percentage of revenue they receive from that customer. If data on such variables is available, future studies on supply networks and innovation would also benefit from incorporating relationship strength and other complementary variables.

Further, while we include R&D intensity as a reflection of a firm's absorptive capacity, there may be other important factors capturing the firm's amount of experience and potential ability to absorb incoming external knowledge. Future research should delve further into other aspects that may affect a firm's ability to absorb knowledge residing in the supply network. Also, we used patent counts as a proxy for innovation output. While prior studies have shown patent counts to be valid and robust indicators of knowledge creation (Trajtenberg, 1987) and highly correlated with citation-weighted patent measures (Hagedoorn and Cloodt, 2003; Stuart, 2000), we acknowledge that additional measures that account for originality and generality as well as type of innovation output are also of importance. An innovation can thus be distinguished based on a backward(forward)-looking measure of originality(generality) (Trajtenberg et al., 1997) or along an innovation continuum of incremental to radical based on the degree of new knowledge embedded in an innovation (Dewar and Dutton, 1986). We do not distinguish between originality, generality, or type of innovation in our study, but future studies accounting for this distinction could aid in help in reducing the current gap between the empirical assessment and conceptualization of firm innovation.

6. Conclusion

In this study, we find further evidence supporting the argument that network structures and relationships that form supply networks are critical components for identifying strategic imperatives in supply chain management (Borgatti and Li, 2009; Kim et al., 2011). Our findings suggest improved knowledge and information flows arising from supply network accessibility influences a firm's innovation output. The results also indicate that interconnected supply networks may help moderate the improved knowledge and information flow from accessibility. Additionally, the results show that the influence of the two structural characteristics on innovation output can be enhanced by a firm's absorptive capacity and level of supply network partner innovativeness. In sum, the study contributes to the body of literature on both supply chain management and innovation by highlighting the role of the structural characteristics of supply networks, along with knowledge variables, in facilitating knowledge creation and thereby improving upon a firm's level of innovation output.

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Appendix A.

2SLS model testing for endogeneity.[a]

Variables	(1) SN accessibility (OLS)	(2) SN interconnectedness (OLS)	(3) Innovation output (2SLS)
Controls			
Firm size	0.00	0.01	0.67***
	(0.02)	(0.01)	(0.06)
Firm age	−0.00*	−0.00	−0.01*
	(0.00)	(0.00)	(0.00)
Industry concentration	−0.15	0.00	−1.67***
	(0.11)	(0.04)	(0.32)
Industry growth[b]	0.75	−0.09	
	(0.39)	(0.18)	
Prior innovation[b]	0.02	−0.02	
	(0.05)	(0.02)	
Prior knowledge breadth	0.00	0.00	0.02***
	(0.00)	(0.00)	(0.00)
Lead firm	0.03	−0.02	0.23
	(0.06)	(0.02)	(0.20)
Degree centrality[b]	0.04***	0.00	
	(0.00)	(0.00)	
Number of pairs[b]	−0.00***	−0.00***	
	(0.00)	(0.00)	
Direct effects			
SN accessibility			0.49**
			(0.18)
SN interconnectedness			0.67
			(1.12)
Absorptive capacity	−0.10	0.02	0.95
	(0.14)	(0.04)	(0.52)
SN partner innovativeness	0.04	0.16**	−0.36
	(0.41)	(0.06)	(0.63)
Constant	0.30**	−0.02	−2.31***
	(0.10)	(0.03)	(0.41)
N	390	390	390

Notes: Models (1) and (2) depict the results for the first-stage regression considering potentially endogenous variables supply network accessibility and supply network interconnectedness, respectively. Model (3) is the resulting 2SLS incorporating the predicted values from the first stage as independent variables to replace the values of the assumed endogenous variables.

[a] Standard errors in parentheses: $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$.

[b] Variables used as instruments for assumed endogenous variables.

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